Natural Language Processing

Natural Language Processing, commonly called NLP, is an area of artificial intelligence focused on enabling computers to process and understand human language. NLP systems work with text or speech and can perform tasks such as classification, translation, summarization, and question answering.

Text must usually be transformed into a numerical representation before a machine learning model can process it. Earlier NLP systems often used methods based on word counts, while modern systems frequently use embeddings that represent semantic meaning.

Tokenization is the process of dividing text into smaller units called tokens. Depending on the model, a token may represent a word, part of a word, punctuation, or another meaningful piece of text.

Embeddings represent words, sentences, or documents as numerical vectors. Text with similar meaning can have similar vector representations, which makes embeddings useful for semantic search, recommendation systems, clustering, and Retrieval-Augmented Generation.

Transformers have become an important architecture in modern NLP. They use attention mechanisms to determine relationships between different parts of an input sequence and can process language more effectively than many earlier sequential architectures.

Large Language Models are commonly trained using transformer architectures and large collections of text. They can generate language, answer questions, summarize documents, extract information, translate text, and assist with software development.

NLP applications still face challenges such as ambiguous language, incorrect model outputs, limited context, biased training data, and domain-specific terminology. Reliable applications often combine language models with retrieval, evaluation, validation, and external tools.